

Yulong Liu

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RESEARCH INTERESTS

Computational Geomechanics | **Poromechanics** | **Reservoir Simulation** | **Scientific Machine Learning**

EDUCATION

Ph.D. in Earth Science – Cornell University 2024 – Present

Minors: Computer Science and Scientific Computing

B.S. in Mining Engineering – Northeastern University (CN) 2024

RESEARCH EXPERIENCE

Ph.D. Student – Dept. of Earth and Atmospheric Sciences, Cornell University Aug. 2024 – Present

Advisor: Chloé Arson

Ithaca, NY

Project: Computational geomechanics and poromechanics for coupled multiphysics problems in subsurface and reservoir systems, supported by physics-based and machine-learning surrogate modeling

- Develop coupled poromechanics and computational geomechanics models of subsurface and reservoir systems (hydraulic fracturing, geothermal injection–production) in the MOOSE finite element framework, capturing thermo-hydro-mechanical interactions in fractured rock.
- Study multiscale transport in porous subsurface media through homogenization theory, informing reservoir-scale surrogate modeling.
- Build physics-informed neural networks, implicit neural representations, and operator-learning surrogates that deliver geometry- and physics-aware predictions at a fraction of the cost of full finite element simulation, supporting engineering decision-making.
- Lead a cross-institution project on PDE-grounded verification of the physical correctness of LLM-generated multiphysics simulation code.
- **Ongoing:** AI-driven discovery of closed-form homogenization laws for permeability—combining data-driven methods with homogenization theory to learn interpretable closed-form expressions for upscaled permeability of heterogeneous porous media.

Undergraduate Researcher – Northeastern University (CN)

Sep. 2020 – Jun. 2024

B.S. in Mining Engineering

Shenyang, China

Thesis: A Unified Model for Microseismic Signal Classification and Arrival-Time Picking Based on Deep Learning

- Conducted research in mining geomechanics spanning hydraulic fracturing, triaxial rock testing, tunnel boring machines (TBMs), and AI-driven microseismic signal processing.
- Developed a unified deep-learning model for microseismic signal classification and arrival-time picking, replacing separate manual processing workflows.
- Performed laboratory rock mechanics testing and supported experimental and field data interpretation for hydraulic fracturing and TBM studies.
- Recognized with the Outstanding Thesis Award for thesis research and the National Scholarship (Ministry of Education) for overall academic performance.

GRANTS, PROPOSALS, AND COMPUTATIONAL AWARDS

Learned Uncertainty-Propagation and World Models for Enhanced Geothermal Reservoirs 2026

National Science Foundation (NSF), ACCESS program. Student lead: Y. Liu; Advisor: C. Arson. Total allocation and Arson group's share: 1.5M CPU hours.

AI-aided Computational Geomechanics

2026

Empire AI Alpha+Beta. PIs: Y.C. Han, Y. Liu, A. Tristani, and C. Arson. Total allocation and Arson group's share: 5.5K GPU hours.

Prototype Groundwater Digital Twin for Aquifer Resilience: Latent World Modeling for Observation-Informed Hydromechanical Prediction 2026

NASA ROSES-25 F.5 FINESST. Under review. Future Investigator: Y. Liu; PI and faculty mentor: C. Arson.

PUBLICATIONS

Selected Papers

- **Yulong Liu**[†], Jonah Botvinick-Greenhouse, Yunan Yang, Chloé Arson. *Operator Learning Surrogate Modeling of Hydraulically Fractured Geothermal Injection–Production Systems: A Cornell Case Study*. ARMA US Rock Mechanics/Geomechanics Symposium, Tucson, AZ, 2026.
- **Liu, Y.**, Arson, C. *A Physics-Informed Neural Network for Pressurized Cavities of Arbitrary Shape in Heterogeneous Rock*. *Rock Mech Rock Eng* (2026). doi:10.1007/s00603-026-05882-5.
- M. Belachew, **Yulong Liu**, J. D. Frost, Chloé Arson. *Numerical Assessment of Plasticity Development and Energy Expenditure of Ant-Like Microtunnelling*. *Tunnelling and Underground Space Technology*, 172, 107501, 2026.
- **Yulong Liu**[†], Chloé Arson. *Physics-Informed Neural Network Surrogate Modeling of Pressurized Cavity in Homogeneous and Bilayered Media*. ARMA US Rock Mechanics/Geomechanics Symposium, D022S018R006, 2025.
- **Yulong Liu**^{*,†}, Zhenghan Song, et al. *Your Simulation Runs but Solves the Wrong Physics: PDE-Grounded Intent Verification for LLM-Generated Multiphysics Simulation Code*. Under review at NeurIPS 2026.
- Zhenghan Song, Yunyi Li, **Yulong Liu**[†]. *Prefix-Safe Bayesian Belief Tracking for LLM Reasoning Reliability: Separating Calibration from Ranking*. Under review at ACL ARR 2026.

Thesis

- **Yulong Liu**. *A Unified Model for Microseismic Signal Classification and Arrival-Time Picking Based on Deep Learning*. Undergraduate thesis, Northeastern University, 2024.

* Co-first author † Corresponding author

CONFERENCES AND PRESENTATIONS

Oral presentation, 60th US Rock Mechanics/Geomechanics Symposium (ARMA) <i>American Rock Mechanics Association, Tucson, AZ, USA</i>	Jun. 2026
Invited talk, 20th Phase Field Methods Workshop <i>Center for Hierarchical Materials Design, Northwestern University, Evanston, IL, USA</i>	May 2026
Poster presentation, Cornell CEE Graduate Research Symposium <i>Cornell University, Ithaca, NY, USA</i>	Apr. 2026
Invited seminar: “Physics-Informed AI Optimization of Heterogeneous Rock Systems” <i>Southwest Petroleum University seminar series, Chengdu, China</i>	Dec. 2025
Poster presentation, 59th US Rock Mechanics/Geomechanics Symposium (ARMA) <i>American Rock Mechanics Association, Santa Fe, NM, USA</i>	Jun. 2025

FELLOWSHIPS AND AWARDS

Estwing Hammer Award – Most Outstanding EAS Graduate Student of 2024, Cornell University	2025
Cornell University Travel Grant , Cornell University	2025, 2026
Outstanding Thesis , Northeastern University	2024
National Scholarship , Ministry of Education, China	2024
First Class Scholarship , Northeastern University	2023
Autumn Scholarship , Northeastern University	2022
Golden Seed Scholarship , Northeastern University	2022

ACADEMIC SERVICE

- Reviewer, *Rock Mechanics and Rock Engineering*
- Reviewer, *Underground Space*
- Reviewer, *International Conference on Artificial Neural Networks*
- Reviewer, *Geothermal Rising Conference*

SKILLS

Programming & Scientific Computing: Python, C++, R, MOOSE finite element framework **Modeling & Simulation:** finite element modeling, coupled multiphysics simulation, poromechanics, computational geomechanics, CAD (e.g., AutoCAD, SolidWorks) **Subsurface & Reservoir:** hydraulic fracturing, geothermal injection–production modeling, multiscale transport in porous media **Scientific Machine Learning:** ML for PDEs, physics-informed neural networks, operator learning, implicit neural representations (INRs), surrogate modeling for scientific computing **LLMs for Simulation:** using large language models to accelerate simulation-software workflows and to verify the physical correctness of generated simulation code **Experimental Methods:** rock mechanics testing (e.g., triaxial compression), laboratory instrumentation **Currently Learning:** phase-field modeling (fracture/multiphysics), multiphase flow in porous media **Scientific Writing:** $\text{\LaTeX}$ / Overleaf

PROFESSIONAL MEMBERSHIPS

- Member, American Rock Mechanics Association (ARMA), 2025–Present
- Member, International Society for Rock Mechanics and Rock Engineering (ISRM), 2025–Present
- Member, Chinese Society for Rock Mechanics and Engineering (CSRME), 2024–Present
- Member, Geothermal Rising, 2025–Present